Factors, Multiples and Patterns — Diagnosing an Overgeneralized Rule
Answer Key

Grades 4–5

Quick Answers

1. 21	3. 4	5. —	7. 29
2. 49	4. 60	6. 9	8. —

Curriculum

Level: Grades 4–5
4.OA.B.4 / 4.OA.C.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

1. *If they got 192: kept doubling from the first term instead of adding three*
2. *If they got 19: kept adding three from the first term, the rule that fits only the first pair*
4. *If they got 115: listed the twelfth number that ends in five instead of the twelfth multiple of five*
7. *If they got 128: kept doubling instead of adding one more each step*

1. Pattern A (3, 6, 9, 12, . . .): test Kip's "double the term before" rule, then find the 7th term.

Step 1: test the rule where Kip stopped looking. Doubling the second term gives $6 \times 2 = 12$, but the third term of Pattern A is 9. The rule fails at the very first term Kip did not check. Step 2: find what really changes from term to term: $6 - 3 = 3$, $9 - 6 = 3$, $12 - 9 = 3$. The step is $+3$ every time, so the nth term is $3 \times n$. Step 3: the 7th term is $3 \times 7 = \boxed{21}$.
Watch for: a student who keeps doubling reaches 192 — six doublings from 3. The size of the gap is the point: an untested rule does not just drift, it runs away.

2. Pattern B (1, 4, 9, 16, . . .): show "add 3" fails, then find the 7th term.

Step 1: Dee's evidence is the single step $1 + 3 = 4$. Test the next step: $4 + 3 = 7$, but the third term is 9. One extra test is all it took. Step 2: name the real pattern — $1 = 1 \times 1$, $4 = 2 \times 2$, $9 = 3 \times 3$, $16 = 4 \times 4$. The nth term is $n \times n$. Step 3: the 7th term is $7 \times 7 = \boxed{49}$.
Watch for: "add 3" carried to the 7th term gives $1 + 3 \times 6 = 19$.

3. Nia: "every multiple of 4 is a multiple of 8". Divide 12 by 8 and give the remainder.

Step 1: check that Ravi's number really is a multiple of 4: $4 \times 3 = 12$ ✓, so 12 belongs to the set Nia's rule is about. Step 2: divide by 8: eight goes into twelve once, and $12 - 8 \times 1$ is what is left over. Step 3: the remainder is $\boxed{4}$. Step 4: a remainder other than zero means 12 is not a multiple of 8, so one counterexample defeats Nia's rule no matter how many supporting examples she has. Her evidence was picked from the multiples of 8 themselves, which is why the rule looked safe.

4. Pattern C (5, 10, 15, 20, . . .): cross out the wrong half of Tam's rule, then find the 12th term.

Step 1: test the ending-in-5 half on the second term. The second term is 10 and it ends in 0, so that half of the rule is false — it was generalized from the first term alone. Cross it out. Step 2: keep the half that survives testing: the terms are the multiples of 5, so the nth term is $5 \times n$. Step 3: the 12th term is $5 \times 12 = \boxed{60}$.

Watch for: a student who keeps the ending-in-5 half lists $5, 15, 25, \ldots$ and answers $5 + 10 \times 11 = 115$.

5. Pattern D (start 4, add 4): compare its 6th term with the 6th term of Bea's doubling rule.

Step 1: both rules agree on the first two terms (4 and 8), which is exactly why Bea's guess survived her own testing. Step 2: Pattern D's 6th term: $4 \times 6 = 24$. Step 3: Bea's rule doubles five times from 4: $4 \times 2^5 = 4 \times 32 = 128$. Step 4: 24 is far less than 128, so the comparison is $\boxed{<}$.

Teaching point: two rules that agree on the first two terms can disagree by over a hundred by the sixth. Agreement early is not evidence.

6. Ollie: "every odd number is prime" (3, 5, 7). Find the smallest odd counterexample greater than 7 and write it as a product.

Step 1: test the odd numbers in order past Ollie's evidence. The next odd number is 9. Step 2: look for a factor pair of 9 other than 1 and 9: 3×3 works, and both factors are smaller than 9 and greater than 1. Step 3: a number with a factor pair like that is composite, not prime, so 9 is a counterexample: $3 \times 3 = \boxed{9}$. Step 4: Ollie's three examples were all prime by coincidence of being small; the first odd square breaks the rule.

7. Pattern E (1, 2, 4, 7, 11, . . .): find where Ines's doubling rule first breaks, then find the 8th term.

Step 1: Ines's evidence covers the first two steps: $1 \times 2 = 2$ ✓ and $2 \times 2 = 4$ ✓. Test the next one: doubling 4 predicts 8, but the fourth term is 7. The rule first breaks at the **4th term**. Step 2: find the real steps: $2 - 1 = 1$, $4 - 2 = 2$, $7 - 4 = 3$, $11 - 7 = 4$. The step grows by one each time. Step 3: continue the steps to the 8th term: after 11 add 5 to get 16, add 6 to get 22, add 7 to get 29. Step 4: check with the shortcut $1 + \frac{n(n-1)}{2}$ at $n = 8$: $1 + \frac{8 \times 7}{2} = 1 + 28 = 29$ ✓. The 8th term is $\boxed{29}$.

Watch for: continued doubling gives $1 \times 2^7 = 128$.

8. Explain: Rosa claims every number ending in 0 is a multiple of 20, citing 20, 40, 60.

Open response — flagged for manual review. Model answer below; accept any correct counterexample and any equivalent reasoning.

(a) Rosa only tested numbers she had already built from 20, so her examples could not disagree with her. Examples can support a rule but never prove it: a rule about *every* number is only proved by an argument that covers all of them, while a single failing case disproves it outright.

(b) Counterexample: 10 ends in 0, and $10 \div 20$ is not a whole number (10 is smaller than 20, so no whole number of 20s fits). Also acceptable: 30 ($30 - 20 = 10$ left over), 50, 70, 90.

(c) A habit a classmate can use: "Before I trust a rule, I test it on a case I did not use to make it — especially one that looks awkward."

What a full-credit answer must contain: a statement that supporting examples are not proof, one genuine counterexample with a reason it fails, and a testing habit stated as a sentence.